GRADUATION PROJECT DOCUMENTATION

Driver Behavior Monitoring System

Real-Time Driver Safety & Compliance Detection Using YOLO & MediaPipe

Submitted in Partial Fulfillment of the Requirements for the

AI Initiative – Graduation Project Program

Project Category	Computer Vision / AI Safety
Technologies	YOLO, MediaPipe, Faster R-CNN, StreamLite
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1. Project Planning & Management

1.1 Project Overview

The Driver Behavior Monitoring System (DBMS) is an AI-powered solution designed to enhance road safety by continuously monitoring driver behavior during vehicle operation. The system leverages state-of-the-art computer vision techniques including YOLO object detection, MediaPipe for facial landmark and body pose analysis, and a Faster R-CNN model for comparison benchmarking.

The system detects critical driver compliance indicators including: seatbelt usage, mobile phone usage, distracted/drowsy behavior, and signs of intoxication. Alerts are issued in real-time via audio notifications when violations are detected. The system supports both live camera streams and pre-recorded video uploads through a Streamlit-based web interface.

1.2 Project Objectives

- Detect and classify driver behavior violations in real-time using YOLO and MediaPipe.
- Implement drowsiness detection using Eye Aspect Ratio (EAR) and head pose analysis via MediaPipe.
- Detect seatbelt compliance using YOLO object detection model.
- Detect mobile phone usage while driving.
- Detect signs of intoxication based on facial/behavioral cues.
- Compare detection performance between YOLO and Faster R-CNN architectures.
- Build a user-friendly Streamlit web application for real-time monitoring and video upload.
- Issue real-time audio alerts upon detecting violations.

1.3 Project Scope

1.3.1 In Scope

- Real-time video stream processing from webcam or IP camera.
- Offline video file analysis and violation detection.
- Driver drowsiness and fatigue detection using MediaPipe.
- Object detection for seatbelt, phone, and alcohol-related cues using YOLO.
- Audio alert generation for detected violations.
- Streamlit dashboard for monitoring and control.
- Model benchmarking: YOLO vs. Faster R-CNN on custom dataset.

1.3.2 Out of Scope

- GPS or vehicle telemetry integration.
- Backend database storage of violation records (future phase).
- Mobile application deployment.
- Integration with law enforcement systems.

1.4 Project Timeline (Gantt Chart Overview)

Phase / Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-10
Requirements & Literature Review	✓	✓			
System Analysis & Design		✓	✓		
Dataset Collection & Labeling		✓	✓		
Model Training (YOLO & Faster R-CNN)			✓	✓	
MediaPipe Integration (Drowsiness)			✓	✓	
Streamlit App Development				✓	✓
Testing, Benchmarking & Documentation				✓	✓

1.5 Team Roles & Responsibilities

Team Member	Responsibilities
Randa	Contributed to AI/ML model development (YOLO & Faster R-CNN), computer vision tasks, and participated in both backend processing and frontend (Streamlit) integration
Yousef	Worked on dataset collection & preprocessing, model training, and contributed to backend development (video streaming pipeline & alert system)
Ahmed	Participated in model training, evaluation, and contributed mainly to backend logic and system pipeline development
Menna	Focused on data annotation, visualization, and contributed to frontend dashboard design and integration
Habiba	Worked on computer vision tasks (MediaPipe, pose estimation) and contributed to frontend and backend integration
Hend	Participated in preprocessing, testing, and contributed to both backend processing and frontend dashboard support

1.6 Risk Assessment & Mitigation Plan

Risk	Probability	Impact	Mitigation
Camera or hardware not available	Medium	Cannot test real-time system	Use pre-recorded videos for testing
Low-light conditions affecting detection	High	Reduced accuracy	Apply brightness and contrast preprocessing
YOLOv8 misclassification (phone/seatbelt)	Medium	Wrong alerts	Improve dataset and adjust confidence threshold
MediaPipe landmark errors	Medium	Wrong drowsiness detection	Use smoothing and frame averaging
Real-time delay (low FPS)	Medium	Not truly real-time	Use lightweight models and optimize processing
Streamlite lag	Low	Poor user experience	Optimize UI updates
Audio alert delay	Low	Late warnings	Use asynchronous alert system
Model overfitting	Medium	Poor generalization	Use data augmentation and validation split
Integration issues (YOLO + MediaPipe)	Low	System instability	Modular testing of each part
Dataset bias	Medium	Unequal accuracy	Use diverse dataset

1.7 Key Performance Indicators (KPIs)

KPI	Target	Measurement Method	Category
Drowsiness detection accuracy	≥ 90%	Test dataset evaluation	Detection
Seatbelt detection accuracy	≥ 85%	YOLO evaluation metrics	Detection
Phone detection precision	≥ 85%	Precision/Recall analysis	Detection
Alert response time	≤ 3 sec	Detection to alert timing	Performance
Real-time speed	≥ 15 FPS	Frame processing test	Performance
System uptime	≥ 99%	Runtime monitoring	Reliability
False alert rate	< 10%	Manual checking	Accuracy
Dashboard response time	≤ 2 sec	Streamlit testing	UX
Model latency	≤ 50 ms/frame	Inference benchmark	Performance

2. Literature Review

2.1 Background

Road traffic accidents remain one of the leading causes of death worldwide, with driver distraction, fatigue, and non-compliance with safety regulations being major contributing factors. According to the World Health Organization (WHO), approximately 1.35 million people die each year due to road traffic crashes. The integration of AI-powered monitoring systems within vehicles presents a promising solution to mitigate these risks through early detection and real-time alerting.

2.2 Object Detection Approaches

2.2.1 YOLO (You Only Look Once)

YOLO is a real-time object detection algorithm introduced by Redmon et al. (2016). Unlike traditional two-stage detectors, YOLO frames object detection as a single regression problem, directly predicting bounding boxes and class probabilities from full images in one evaluation. This gives YOLO a significant speed advantage, making it ideal for real-time applications such as vehicle monitoring systems.

YOLOv8 (Ultralytics, 2023) is the latest iteration, offering improved accuracy/speed trade-offs, anchor-free detection, and native support for Python-based deployment. The model will be fine-tuned on a custom dataset for seatbelt, mobile phone, and intoxication-related object detection.

2.2.2 Faster R-CNN

Faster R-CNN (Ren et al., 2015) introduced the Region Proposal Network (RPN) which generates candidate object regions, subsequently classified and refined by a Fast R-CNN network. While significantly more accurate on complex scenes than single-stage detectors, Faster R-CNN typically requires more computation time, making it less suitable for strict real-time applications but valuable as a benchmarking baseline.

In this project, Faster R-CNN will be trained on the same custom dataset as YOLO to provide a direct performance comparison in terms of mAP (mean Average Precision), inference time, and resource utilization.

2.3 Facial Landmark & Pose Estimation

2.3.1 MediaPipe

MediaPipe (Google, 2019) is an open-source framework for building cross-platform, customizable ML pipelines. The FaceMesh and Pose modules provide real-time detection of 468 3D facial landmarks and full-body pose estimation respectively. For driver monitoring, the following capabilities are leveraged:

- Eye Aspect Ratio (EAR): Calculated from eye landmarks to determine blink frequency and duration, a primary indicator of drowsiness.
- Head Pose Estimation: Yaw, pitch, and roll angles derived from facial landmarks detect head nodding and prolonged looking-away behavior.
- Mouth Aspect Ratio (MAR): Detects yawning patterns associated with fatigue

2.4 Drowsiness Detection Methods

Several approaches have been documented in literature for drowsiness detection. Traditional methods relied on steering behavior analysis or lane deviation. Modern approaches use vision-based methods: PERCLOS (Percentage of Eye Closure), head movement analysis, and facial action unit detection. This system combines EAR thresholding with head pose analysis through MediaPipe for a multi-modal, robust detection approach.

2.5 Streamlit for AI Applications

Streamlit is an open-source Python framework that enables rapid development of web applications for machine learning and data science. Its simplicity allows AI teams to deploy functional interfaces without front-end expertise, making it ideal for prototype and demonstration-phase AI systems such as DBMS.

2.6 Related Work Comparison

Study/System	Detection Focus	Method	Real-Time	Limitation
Owayjan et al. (2019)	Drowsiness	EAR + SVM	Yes	No seatbelt/phone detection
Zhao et al. (2020)	Distraction	CNN	Partial	No audio alert system
Ali et al. (2021)	Seatbelt + Phone	Faster R-CNN	No	Speed too slow
Proposed DBMS	Multi-behavior	YOLO + MediaPipe	Yes	Early prototype stage

2.7 Feedback & Evaluation Criteria

Evaluation Area	Weight	Assessment Criteria
System Documentation	20%	Completeness of report, clarity of explanation, UML diagrams, system design quality
Implementation Quality	30%	Code structure, modular design, error handling, integration of YOLO & MediaPipe, production readiness
Testing & Validation	20%	Model evaluation (accuracy, precision, recall), real-time testing, performance metrics
Innovation & Impact	15%	Novelty of using AI for driver monitoring, real-world applicability, multi-behavior detection
Presentation & Demo	15%	Quality of live demonstration, clarity of slides, ability to answer questions

2.8 Suggested Improvements

To enhance the performance and efficiency of the Driver Behavior Monitoring System in

real-world environments, several future improvements are proposed to improve accuracy, reliability, and scalability:

- Improve system performance in low-light conditions by using infrared-enabled cameras or real-time image enhancement techniques
- Develop a gaze estimation module to more accurately track the driver's focus direction instead of relying only on head movement
- Integrate physiological signals from wearable devices (e.g., heart rate) to enable multi-modal fatigue detection
- Apply per-user calibration for the Eye Aspect Ratio (EAR) to improve accuracy and reduce errors caused by individual differences
- Implement an automated model retraining pipeline using MLOps tools such as MLflow to ensure continuous performance improvement
- Extend the detection system to include additional behaviors such as eating, drinking, or using hands for distracting activities
- Add a Text-to-Speech (TTS) based voice alert system alongside traditional alarms to improve warning effectiveness
- Improve system scalability by supporting multiple in-car cameras simultaneously to increase monitoring coverage

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Role	Primary Concern
Traffic Safety Authority	System Commissioner	Reduce accident rates
Fleet Manager / Transport Co.	Primary User	Monitor driver compliance in real-time
Driver	Subject of Monitoring	Fair and accurate detection
AI Development Team	Builder	Accurate models, maintainable codebase
Ministry / Initiative	Evaluator	Functional prototype, proper documentation

3.2 Functional Requirements

FR-01: Object Detection

- The system SHALL detect whether the driver is wearing a seatbelt.
- The system SHALL detect mobile phone usage by the driver.
- The system SHALL detect objects or behaviors indicating intoxication.
- Detection SHALL occur in real-time (≥15 FPS on standard hardware).

FR-02: Drowsiness & Alertness Detection

- The system SHALL monitor driver eye state using MediaPipe FaceMesh.
- The system SHALL calculate Eye Aspect Ratio (EAR) and issue alerts when EAR falls below threshold for a defined duration.
- The system SHALL monitor head pose for prolonged downward or sideways tilt.
- The system SHALL detect yawning via Mouth Aspect Ratio (MAR).

FR-03: Alert System

- The system SHALL issue an audio alert within 2 seconds of detecting a violation.
- Each violation type SHALL have a distinct audio alert message.
- Alerts SHALL continue until the violation condition is resolved.

FR-04: Input Modes

- The system SHALL accept live video stream from webcam or IP camera via Streamlit.
- The system SHALL accept pre-recorded video file uploads for analysis.

FR-05: Model Comparison

- The system SHALL train both YOLO and Faster R-CNN on the same dataset.
- A benchmarking report SHALL compare mAP, FPS, precision, and recall.

3.3 Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	System must process video at minimum 15 FPS for real-time detection
NFR-02	Accuracy	YOLO model mAP@0.5 must exceed 80% on test dataset
NFR-03	Usability	Streamlit interface must be operable without technical training
NFR-04	Reliability	System uptime must exceed 99% during active monitoring sessions
NFR-05	Portability	System must run on standard laptop hardware (GPU optional)
NFR-06	Latency	Alert must trigger within 2 seconds of violation detection

4. System Analysis & Design

4.1 System Architecture Overview

The Driver Behavior Monitoring System follows a modular pipeline architecture composed of five main layers: Input Layer, Processing Layer, Detection Layer, Alert Layer, and Presentation Layer. The system is designed for extensibility, allowing independent upgrades to individual modules.

4.1.1 System Architecture Diagram

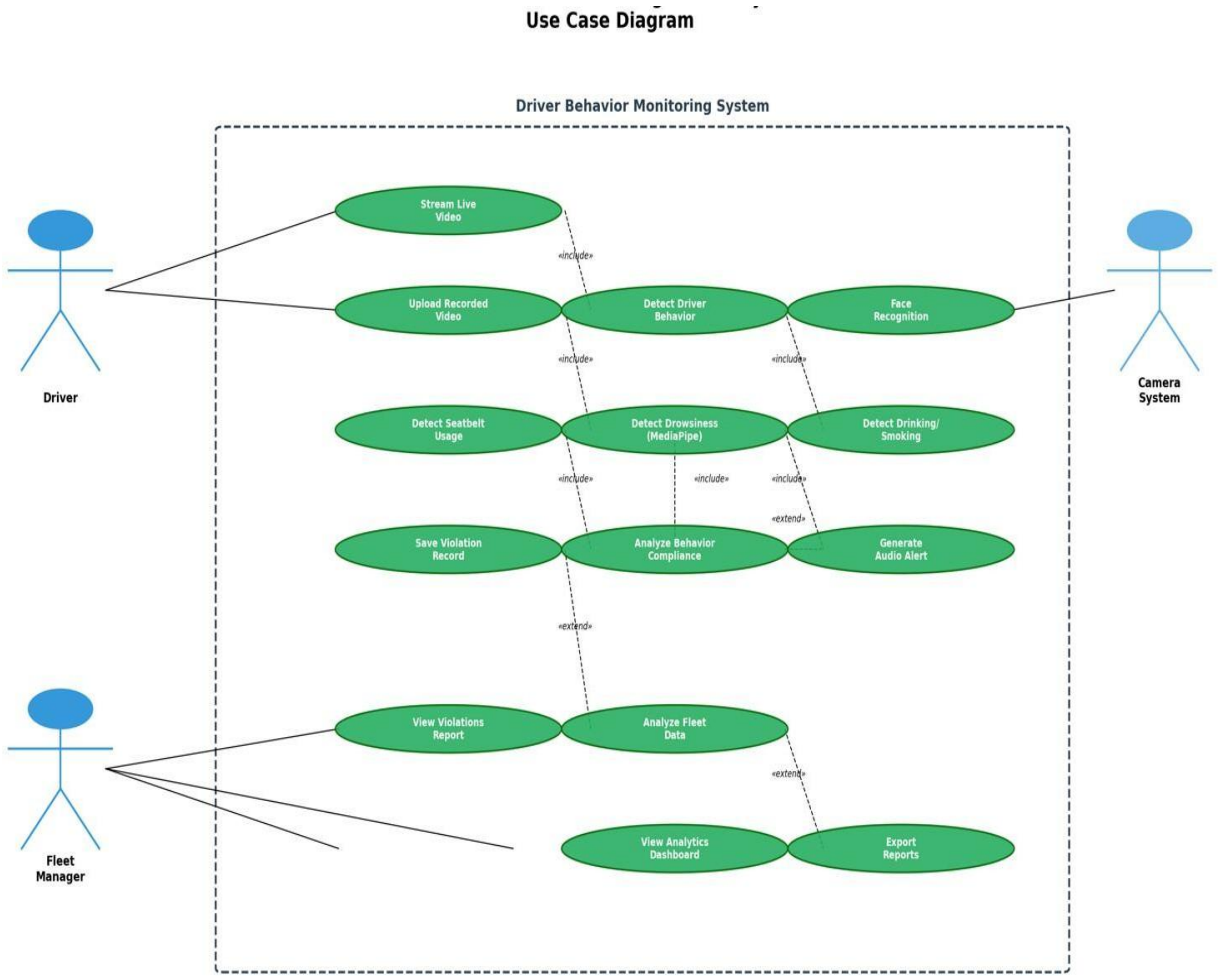
```
DRIVER BEHAVIOR MONITORING SYSTEM — ARCHITECTURE
INPUT LAYER        [ Webcam / IP Camera Stream ]      [ Uploaded Video File ]
                   ↓                                                  ↓
PROCESSING LAYER   [ Frame Extraction ]  →  [ Pre-processing: Resize, Normalize ]
                   ↓                                  ↓
DETECTION LAYER    [ YOLO Detector ]          [ MediaPipe FaceMesh & Pose ]
                   [ Seatbelt | Phone | DUI ]      [ EAR | Head Pose | MAR ]
                   ↓                                  ↓
DECISION ENGINE    [ Violation Logic: Threshold Check + Duration Filter ]
                                          ↓
ALERT LAYER        [ Audio Alert Generator ]  [ Violation Log ]
                                          ↓
PRESENTATION LAYER [ Streamlit Dashboard: Live View + Stats + Controls ]
```

4.2 Use Case Diagram

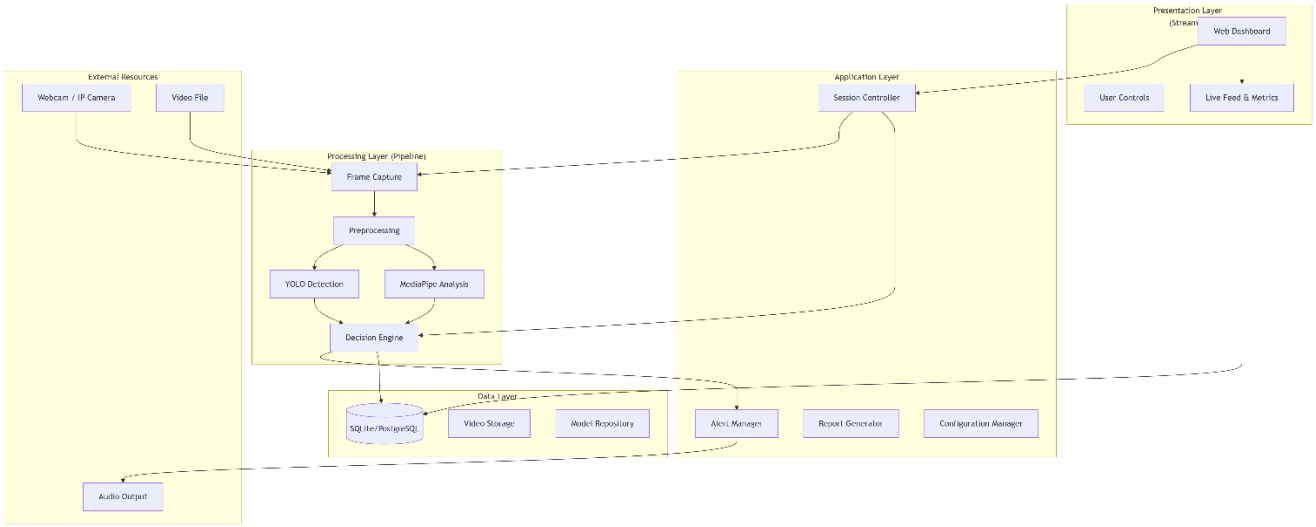
4.2.1 Actors and Use Cases

Actor	Use Cases
Fleet Manager / Operator	Start Monitoring Session, Upload Video, View Live Feed, Receive Alert, View Violation Report
System (AI Engine)	Detect Seatbelt, Detect Phone Usage, Detect Drowsiness, Detect Intoxication, Generate Alert, Log Violation
Driver	Subject of monitoring (passive actor)

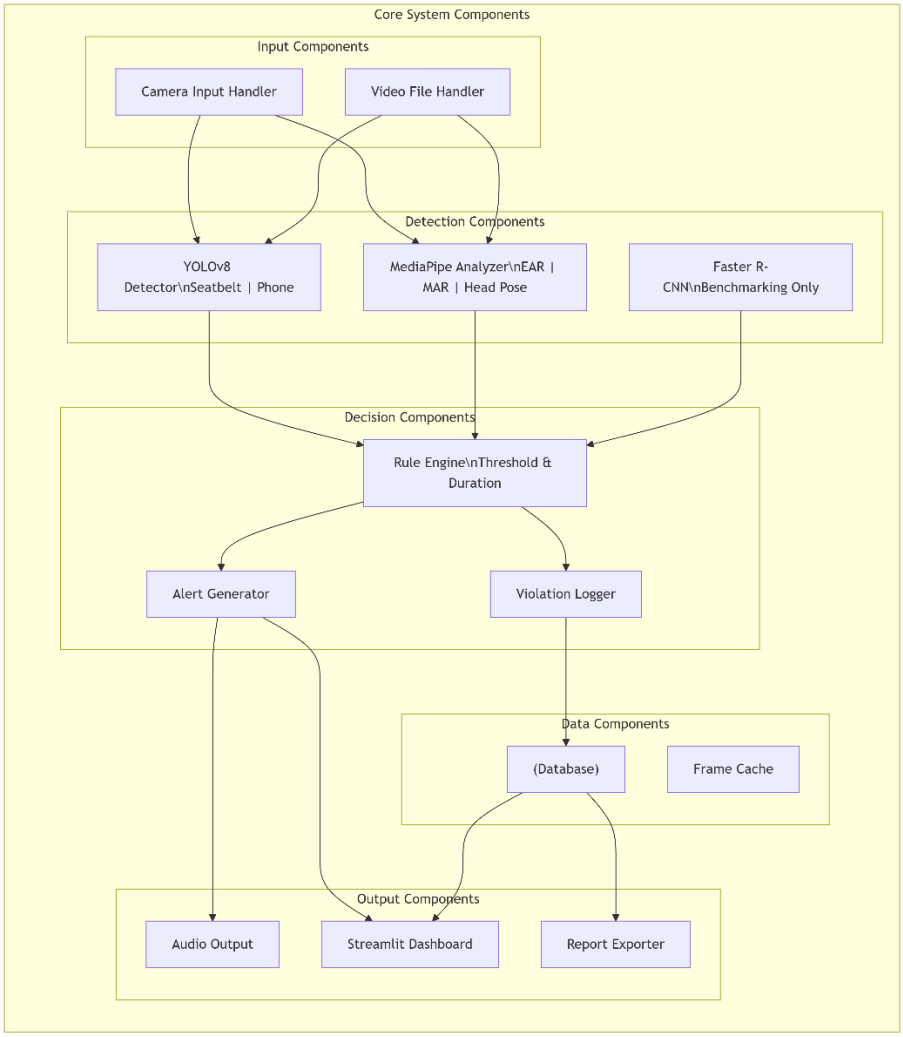
4.2.2 Use Case Diagram (Text Representation)



4.3 Software Architecture



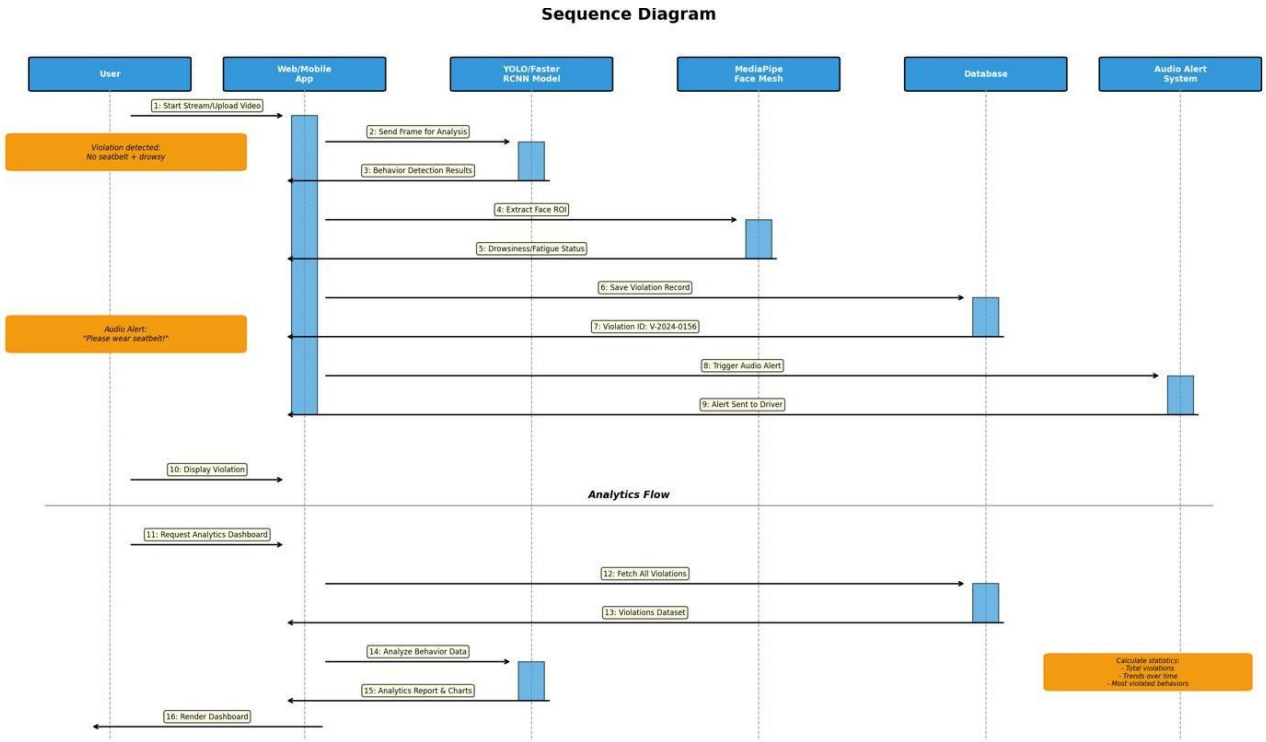
4.4 Component Architecture Diagram



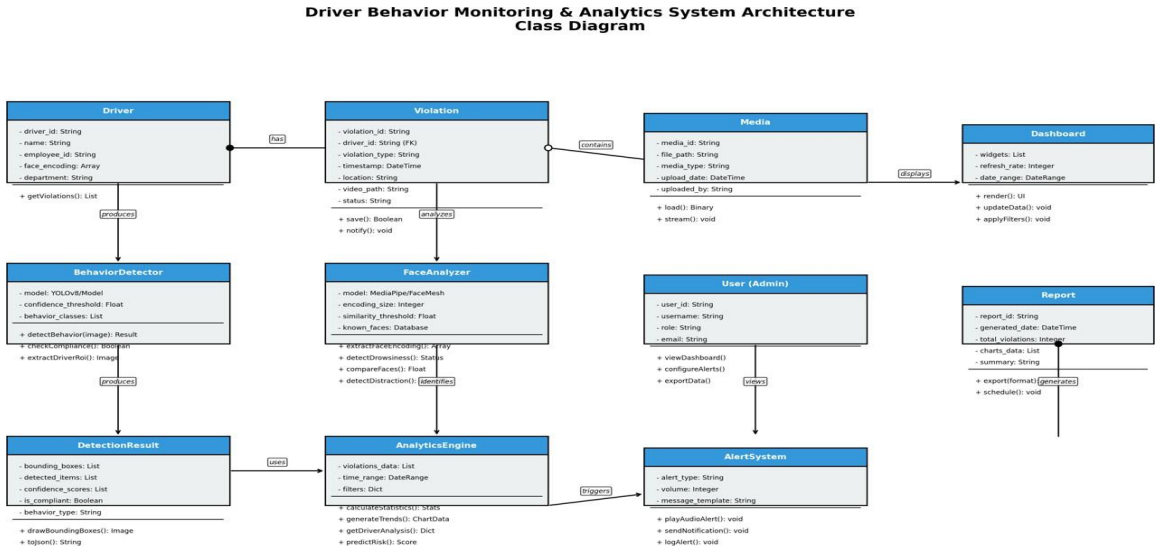
Data Flow & System Behavior

4.5 Sequence Diagrams

4.5.1 Sequence Diagram:

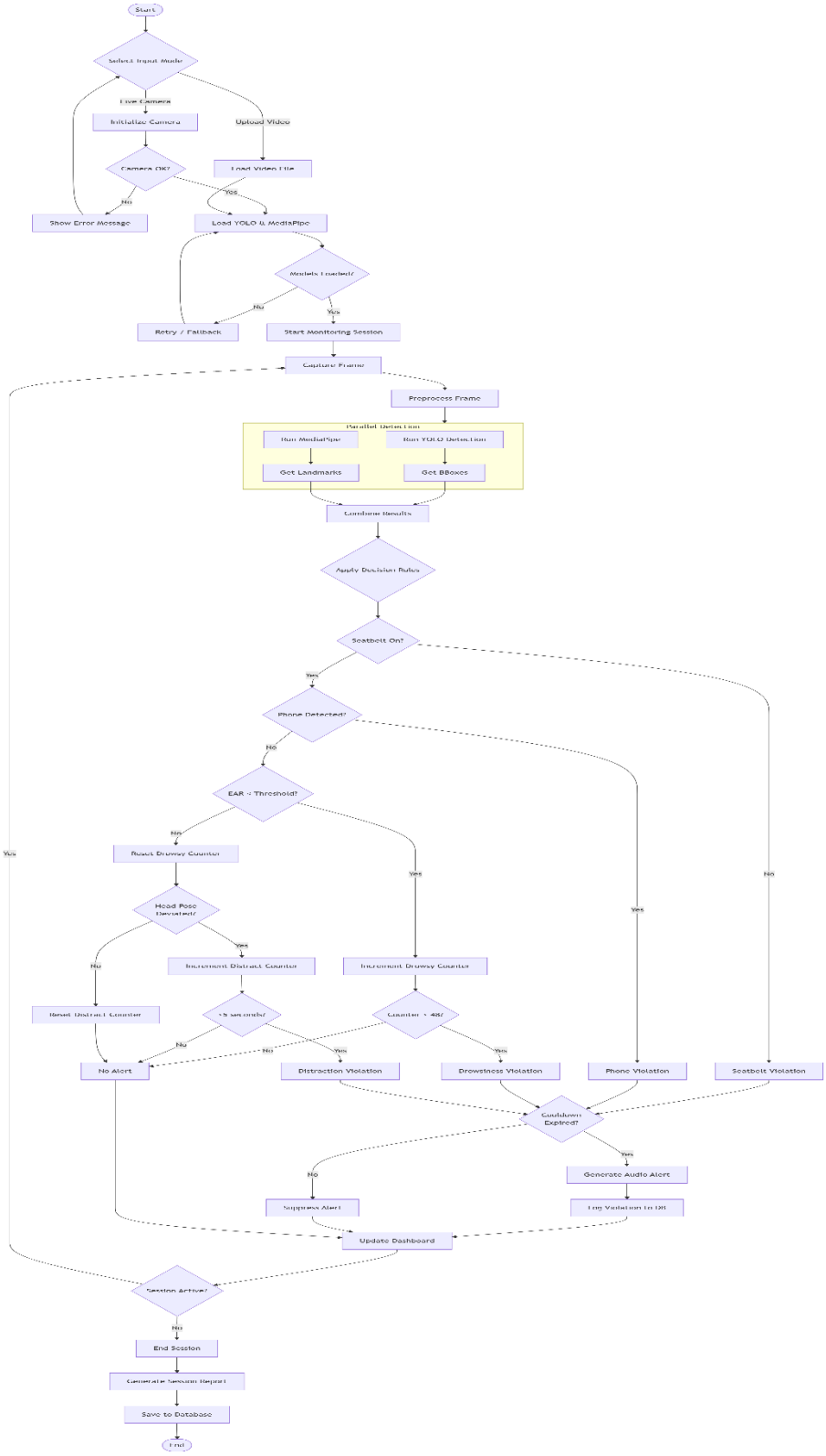


4.6 Class Diagram

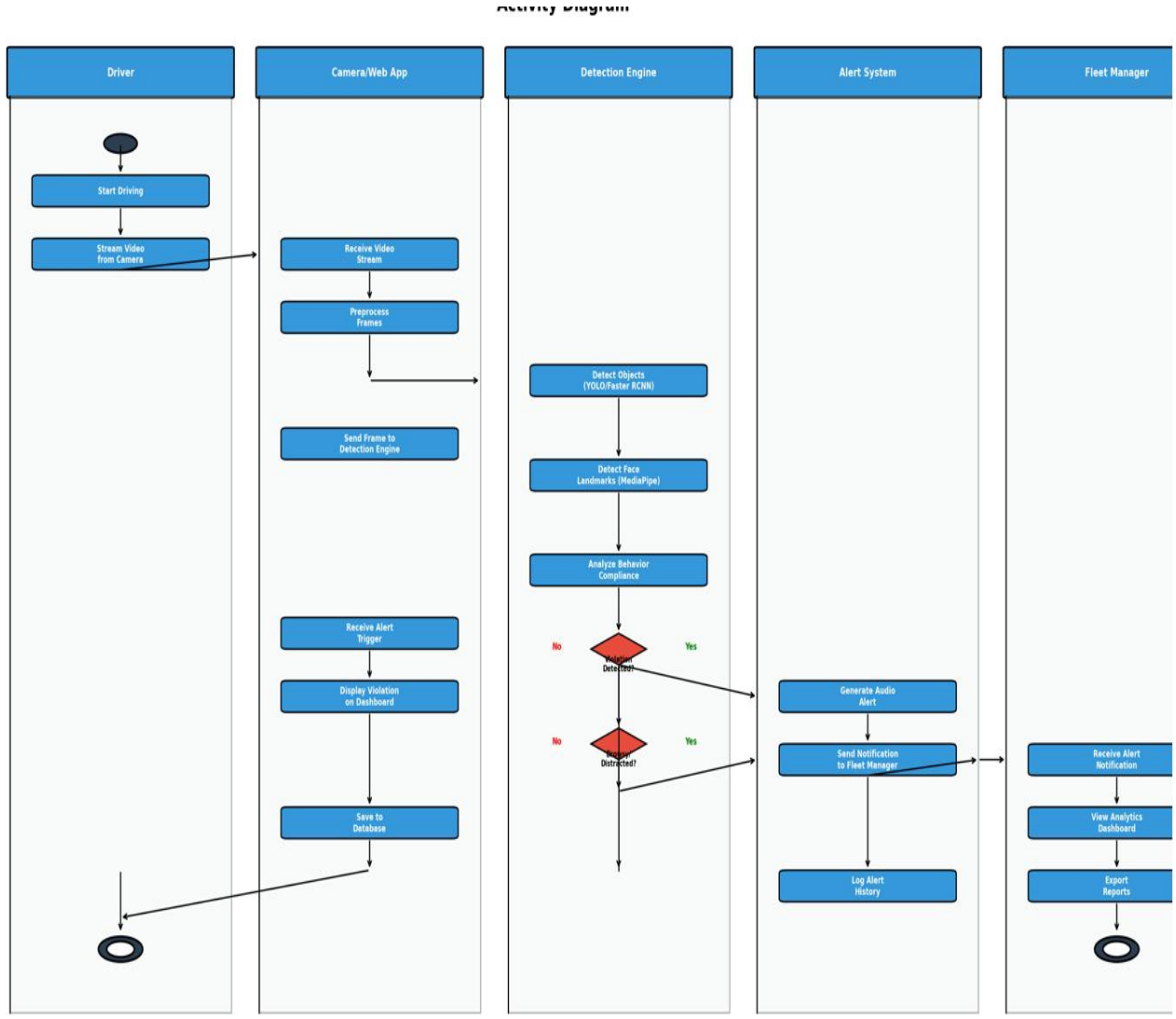


4.7 Activity Diagram:

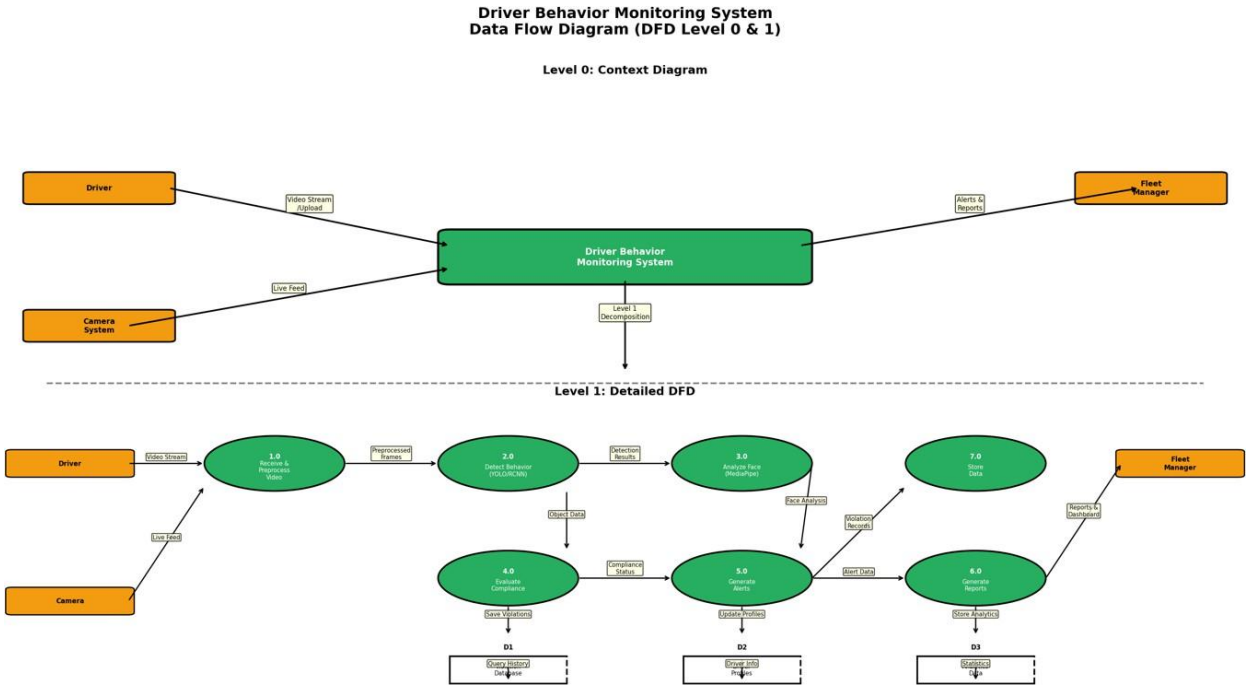
4.7.1 Complete System Activity Diagram



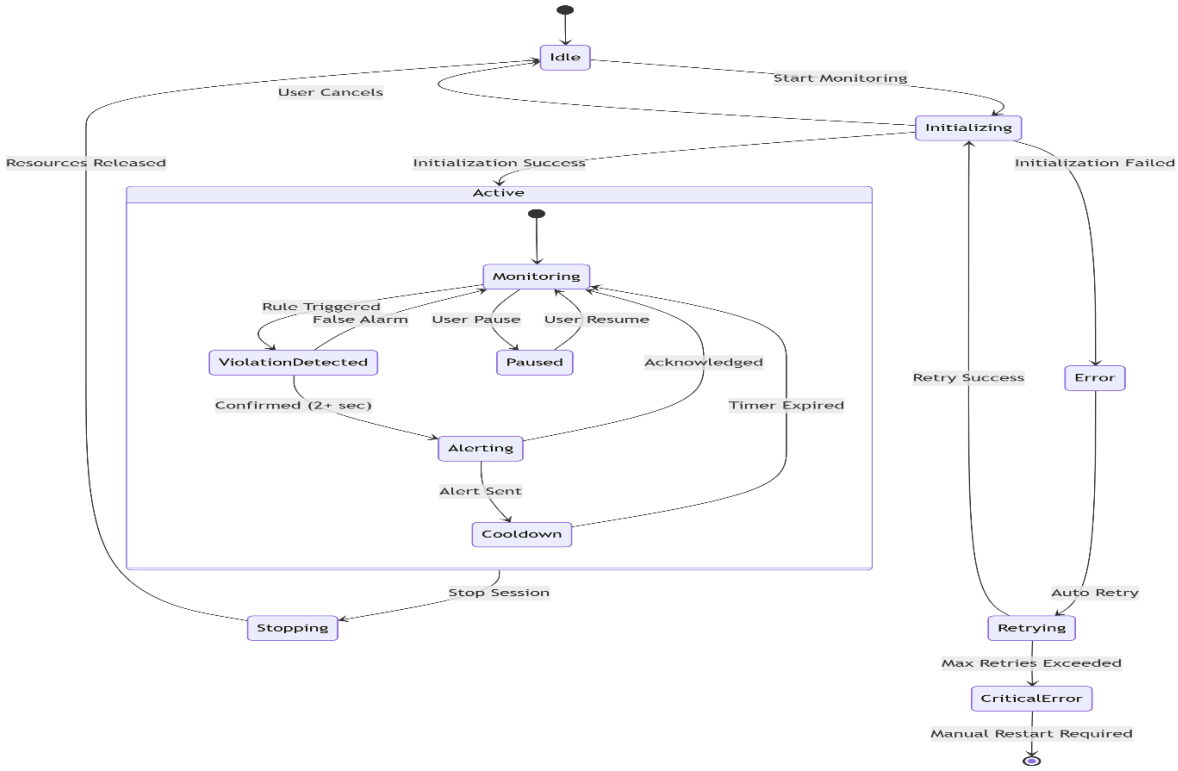
4.7.2 Violation Detection Flow



4.8 Data Flow Diagram (DFD)



4.9 State Diagram



4.10 Model Benchmarking Plan

Both YOLO (YOLOv8) and Faster R-CNN will be trained and evaluated on the same custom annotated dataset. The following metrics will be compared:

Metric	YOLO (Expected)	Faster R-CNN (Expected)
Inference Speed (FPS)	> 30 FPS (Real-time)	5-10 FPS (Slow)
mAP@0.5	~85%	~88%
Precision	~82%	~86%
Recall	~80%	~84%
Model Size	~6 MB (YOLOv8n)	~150 MB
Hardware Requirement	CPU feasible	GPU recommended
Deployment Suitability	✓ Real-time	✗ Too slow for real-time

Based on the benchmarking results, YOLO will be selected as the primary production model for the DBMS due to its real-time inference capability, compact model size, and sufficient accuracy for the target deployment environment

5. Technology Stack

Component	Technology	Purpose
Object Detection	YOLOv8 (Ultralytics)	Primary real-time detection of seatbelt, phone, intoxication
Benchmarking	Faster R-CNN (PyTorch)	Model comparison for accuracy vs. speed trade-off
Facial Analysis	MediaPipe FaceMesh	Eye aspect ratio, yawning, head pose estimation
Pose Estimation	MediaPipe Pose	Body posture and driver alertness indicators
Frontend / UI	Streamlit	Web-based dashboard for live monitoring and uploads
Backend / Logic	Python 3.10+	Core processing pipeline and integration layer
Audio Alerts	pyttsx3 / playsound	Text-to-speech and audio alert generation
Video Processing	OpenCV (cv2)	Frame capture, annotation, video I/O
Deep Learning	PyTorch / TensorFlow	Model training and inference backend
Data Annotation	LabelImg / Roboflow	Dataset labeling for YOLO and R-CNN training
Version Control	Git / GitHub	Source code and documentation management

6. References

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